

Growing Together

Community Afforestation & Tree Health System

CSC 236: Object Oriented Programming 2 Project

Imam Abdulrahman bin Faisal University



GROUP TEAM MEMBERS

#	Student name	Student ID	
1	Shaden Dhaifallh Alghamdi	2250004960	Section 1
2	Fatimah Ahmed Altaher	2250002084	Section 1
3	Razan Hafiz Sahli	2250005885	Section 1
4	Fajr Hamad Alassaf	2250000194	Section 1

PROBLEM STATEMENT

- **Environmental Challenges:** Rapid urban growth lacks structured greenery management and centralized coordination.
- **High Mortality Rates:** Planted trees often fail over time due to a complete lack of systematic health monitoring.
- **Inefficient Coordination:** Absence of a unified digital tool to sync community volunteer efforts with strategic campaign goals.

SYSTEM OBJECTIVES

Structured Campaigns

Establishing clear afforestation goals by allowing supervisors to seamlessly manage event locations, dates, and target tree counts within an organized digital framework.

Longevity Tracking

Ensuring sustainable tree survival rates by logging periodic health updates and tracking conditions like "Healthy," "Dead," or "Not Seen" across seasons.

CORE FUNCTIONALITIES

Afforestation

Direct implementation of planting sessions and instant campaign target reach calculation.

Volunteers

Simplified digital management, recruitment, and task assignment of project volunteers.

Monitoring

Periodic health status logging and historic progress records for long-term tree maintenance.

SUPERVISOR CONTROLS

Campaign Management

Complete authority over selecting strategic parameters, including exact geographic location, target dates, and ecological milestone goals.

Data Documentation

Accurate logging of specific tree species, amounts, and positions to safeguard data integrity and feed long-term health metrics.

TECHNICAL CHALLENGES

- **Logic Implementation:** Building a robust hierarchical OOP structure to connect complex entities like Trees, Campaigns, and Volunteers.
- **GUI Navigation:** Designing fluid, thread-safe transitions between supervisor configuration dashboards and record tracking logs.
- **Data Lifecycle:** Handling data persistence layers across multiple tracking sessions effectively without losing status updates.

RESOLUTIONS & SOLUTIONS

Encapsulation & OOP

Successfully separated backend business logic components from front-end GUI components to achieve highly modular and robust code design.

Java Collections API

Utilized highly efficient framework collections (like ArrayLists and tracking Maps) to handle real-time student, volunteer, and tree queries quickly.

THANK YOU

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